

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **Python**-based data pipelines for ETL processes, transforming and loading data from disparate sources into a unified data warehouse, processing approximately **1.5TB** of data weekly.
- Engineered **Java Spring Boot** microservices to build and expose REST APIs for core business functionalities, handling over **5,000 requests per minute** during peak hours.
- Implemented **FastAPI** inference endpoints for machine learning models, achieving an average **p95 latency of 180ms** for real-time predictions on incoming data streams.
- Developed and deployed **Generative AI solutions**, including implementing **prompt engineering** strategies and building **RAG frameworks** to enhance information retrieval accuracy.
- Constructed evaluation frameworks for **LLMs** to measure efficacy and guide product development, defining key performance indicators and ground truth dataset quality metrics.
- Collaborated with cross-functional teams, including **product managers** and **business leaders**, to translate complex requirements into actionable technical designs and deliverables.
- Containerized AI services using **Docker** and orchestrated deployments on **Kubernetes** clusters, ensuring scalable and resilient production environments.

SKILLS

AI/ML – Generative AI, LLMs, RAG, Prompt Engineering, scikit-learn, Machine Learning Fundamentals, NLP

Languages – Python, Java, SQL, R

Frameworks/Libraries – Spring Boot, FastAPI, Flask, pandas, NumPy, LangChain

Databases – PostgreSQL, MySQL, Oracle, Vector Databases

Cloud/DevOps – AWS, Docker, Kubernetes, CI/CD, Jenkins, Git

BI/Analytics – Power BI, Tableau, DAX, Power Query, Data Warehousing, Statistical Analysis

PROJECTS

Generative AI-Powered Document Summarization Service

- Developed a **Python** service using **FastAPI** to ingest large documents and generate concise summaries via integration with a large language model.
- Implemented a **RAG framework** with a **vector database** to ground LLM responses in specific organizational knowledge bases, improving factual accuracy by an estimated **30%**.
- Containerized the application using **Docker** for consistent deployment across development and production environments.

Customer Transaction Anomaly Detection Model

- Engineered a **Python** feature pipeline to extract behavioural signals from transaction logs, processing approximately **500,000 records** to prepare data for model training.
- Trained a **scikit-learn** classifier to detect anomalous transactions, achieving a **0.88 AUC** and **0.42 F1-score** on a held-out dataset.
- Deployed the trained model as a REST API endpoint, enabling real-time scoring of new transactions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)